

Port City International University

9R55+PP8, S Khulshi Rd, Chattogram

Lab Report : Final Term (Spring-24 -Trimester)

CNN for Image Classification by CIFAR-10 Dataset

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Submitted By:

Name: Mizanur Rahman

Id: CSE -02407232

Batch : 24D-A

Trimester: spring 24

Submitted To :

Manoara Begum

Assistant Professor

Department of CSE,PCIU

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Introduction

The CIFAR-10 dataset is a benchmark dataset widely used in the field of computer vision for image classification tasks. It comprises 60,000 32x32 color images in 10 classes, with each class containing 6,000 images. The dataset is divided into a training set of 50,000 images and a testing set of 10,000 images. Each image is labeled with one of the following classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, or truck. The dataset is commonly employed to evaluate the performance of machine learning algorithms and deep learning models for image classification.

Dataset Description

Training Set: Contains 50,000 images, each labeled with one of the 10 classes. This set is utilized to train machine learning models.

Testing Set: Consists of 10,000 images, also labeled with one of the 10 classes. It serves as a benchmark to evaluate the performance of trained models on unseen data.

Image Size: Each image in the dataset is a 32x32 pixel RGB image, meaning it has three color channels (Red, Green, Blue).

Classes: The dataset is divided into 10 classes, with each class representing a distinct object category.

Label Distribution: The dataset is evenly distributed across the classes, with each class containing an equal number of images.

Experimental Setup

To demonstrate the use of the CIFAR-10 dataset for image classification, we will develop a Convolutional Neural Network (CNN) using Python and TensorFlow/Keras libraries. The CNN will be trained on the training set and evaluated on the testing set. We will follow these steps:

- Import necessary libraries.
- Load and preprocess the CIFAR-10 dataset.
- Define the CNN architecture.

- Compile the model with appropriate loss function and optimizer.
- Train the model on the training data.
- Evaluate the model on the testing data.
- Analyze the performance of the model.

Code

Import Libraries

```
import tensorflow as tf
from tensorflow.keras import layers, models, datasets
```

Function to resize images

```
def resize_images(images, size):
    resized_images = tf.image.resize(images, size)
    return resized_images
```

Load and preprocess the CIFAR-10 dataset

```
(train_images, train_labels), (test_images, test_labels) =
datasets.cifar10.load_data()

# Change the data format from (height, width, channels) to (batch,
height, width, channels)
train_images = train_images.reshape(-1, 32, 32, 3)
test_images = test_images.reshape(-1, 32, 32, 3)

train_images = resize_images(train_images, (28, 28))
```

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```
test_images = resize_images(test_images, (28, 28))  
  
train_images, test_images = train_images / 255.0, test_images / 255.0
```

Define the CNN architecture

```
model = models.Sequential([  
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28,  
3)),  
    layers.MaxPooling2D((2, 2)),  
    layers.Conv2D(64, (3, 3), activation='relu'),  
    layers.MaxPooling2D((2, 2)),  
    layers.Conv2D(64, (3, 3), activation='relu'),  
    layers.Flatten(),  
    layers.Dense(64, activation='relu'),  
    layers.Dense(10)  
])
```

Compile the model

```
model.compile(optimizer='adam',  
  
loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),  
            metrics=['accuracy'])
```

Train the model

```
history = model.fit(train_images, train_labels, epochs=10,  
                    validation_data=(test_images, test_labels))
```

Epoch 1/10

```

1563/1563 _____ 23s 14ms/step - accuracy: 0.3374 -
loss: 1.7952 - val_accuracy: 0.5132 - val_loss: 1.3378
Epoch 2/10
1563/1563 _____ 26s 16ms/step - accuracy: 0.5374 -
loss: 1.2859 - val_accuracy: 0.5389 - val_loss: 1.2568
Epoch 3/10
1563/1563 _____ 37s 14ms/step - accuracy: 0.5974 -
loss: 1.1310 - val_accuracy: 0.6079 - val_loss: 1.0947
Epoch 4/10
1563/1563 _____ 41s 14ms/step - accuracy: 0.6368 -
loss: 1.0302 - val_accuracy: 0.6430 - val_loss: 1.0241
Epoch 5/10
1563/1563 _____ 47s 18ms/step - accuracy: 0.6670 -
loss: 0.9548 - val_accuracy: 0.6046 - val_loss: 1.1310
Epoch 6/10
1563/1563 _____ 22s 14ms/step - accuracy: 0.6820 -
loss: 0.9067 - val_accuracy: 0.6437 - val_loss: 1.0366
Epoch 7/10
1563/1563 _____ 21s 14ms/step - accuracy: 0.7029 -
loss: 0.8445 - val_accuracy: 0.6653 - val_loss: 0.9776
Epoch 8/10
1563/1563 _____ 20s 13ms/step - accuracy: 0.7188 -
loss: 0.8001 - val_accuracy: 0.6627 - val_loss: 0.9782
Epoch 9/10
1563/1563 _____ 20s 13ms/step - accuracy: 0.7319 -
loss: 0.7655 - val_accuracy: 0.6683 - val_loss: 0.9557
Epoch 10/10
1563/1563 _____ 22s 14ms/step - accuracy: 0.7473 -
loss: 0.7239 - val_accuracy: 0.6653 - val_loss: 0.9780

```

Evaluate the model

```

test_loss, test_acc = model.evaluate(test_images, test_labels,
verbose=2)
print('\nTest accuracy:', test_acc)

```

```

313/313 - 1s - 4ms/step - accuracy: 0.6653 - loss: 0.9780

```

```

Test accuracy: 0.6653000116348267

```

Explanation

We begin by importing the necessary libraries, including TensorFlow and Keras. Next, we load the CIFAR-10 dataset using the `datasets.cifar10.load_data()` function and preprocess the images by scaling their pixel values to the range $[0, 1]$. We then define a CNN architecture using the Sequential API of Keras, consisting of convolutional layers, max-pooling layers, and dense layers. The model is compiled with the Adam optimizer and sparse categorical cross-entropy loss function. The model is trained on the training data for 10 epochs with validation data provided for monitoring. Finally, we evaluate the trained model on the testing data and print the test accuracy.

Conclusion

In this lab report, we have explored the CIFAR-10 dataset, which is a widely used benchmark dataset for image classification tasks. We have demonstrated the process of building and training a Convolutional Neural Network using Python and TensorFlow/Keras to classify images from the CIFAR-10 dataset. The trained model achieved a test accuracy that serves as a measure of its performance on unseen data. This experiment showcases the application of deep learning techniques for image classification tasks using real-world datasets like CIFAR-10.